Exercises 7.4 1, 3, 5, 7, 11, 15, 17, 19, 21, 23, 27, 29, 31, 33, 35, 37, 39, 41, 47, 49, 51, 65.

1. (a) $\dfrac{1}{(x-3)(x+5)}$

$= \dfrac{A}{x-3} + \dfrac{B}{x+5}$

2. (a) $\dfrac{x-6}{x^2+x-6}$

$= \dfrac{1}{(x+3)(x-2)} = \dfrac{A}{x+3} + \dfrac{B}{x-2}$

3. (a) $\dfrac{x^2+4}{x^3-3x^2+2x}$

$= \dfrac{x^2+4}{x(x-2)(x-1)} = \dfrac{A}{x} + \dfrac{B}{x-2} + \dfrac{C}{x-1}$

4. (a) $\dfrac{5}{x^4-1}$

$= \dfrac{5}{(x^2+1)(x+1)(x-1)}$

$= \dfrac{A}{x+1} + \dfrac{B}{x-1} + \dfrac{Cx+D}{x^2+1}$

5. (a) $\dfrac{x^5+1}{(x^2-x)(x^4+2x^2+1)}$

$= \dfrac{x^5+1}{x(x-1)(x^2+1)^2}$

$= \dfrac{A}{x} + \dfrac{B}{x-1} + \dfrac{C}{x^2+1} + \dfrac{Dx+E}{(x^2+1)^2}$

15. $\int \frac{x^2}{x-1}\,dx$

$= \int x+1+\frac{1}{x-1}\,dx$

$= \frac{x^2}{2} + x + \ln|x-1|$

$$\begin{array}{r} x+1 \\ x-1 \overline{)\,x^2+0x+0} \\ \underline{x^2-x} \\ x+0 \\ \underline{x-1} \\ 1 \end{array}$$

19. $\int_0^1 \frac{x^2+x+1}{(x+1)^2(x+2)}\,dx$

$= \int_0^1 \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}\,dx$

$x^2+x+1 = A(x+1)(x+2) + B(x+2) + C(x+1)^2$

21. $\int \frac{dt}{(t^2-1)^2} = \int \frac{dt}{[(t+1)(t-1)]^2}$

$= \int \frac{A}{t^2-1} + \frac{B}{(t^2-1)^2}\, dx = \int \frac{dt}{(t+1)^2(t-1)^2}$

$1 = A(t^2-1) + B$

$= \int \frac{A}{t+1} + \frac{B}{(t+1)^2} + \frac{C}{t-1} + \frac{D}{(t-1)^2}\, dx$

① $t = 1$

$B = 1$

② $t = 0$

$1 = A + 1$

$A = 0$

$= \int \frac{1}{(t^2-1)^2}\, dx$?

let $u = t^2-1$; $\frac{du}{dx} = 2t \Rightarrow dx = \frac{du}{2t}$

$t = \sqrt{u-1}$

$= \frac{1}{2}\int \frac{du}{u^2\sqrt{u-1}}$

23. $\int \frac{10}{(x-1)(x^2+9)}\,dx$

$= \int \frac{A}{x-1} + \frac{B}{x^2+9} + \frac{Cx+D}{x^2+9}\,dx$

27. $\int \dfrac{4x}{x^3 + x^2 + x + 1}\, dx$

$$
\begin{array}{r|l}
x+1 & x^3 + x^2 + x + 1 \\
 & x^3 + x^2 \\
\hline
 & 0 + x \\
 & 0 \quad 0 \\
\hline
 & x + 1 \\
 & x + 1 \\
\hline
 & 0
\end{array}
\qquad x^2 \;\; 0 \;\; 1
$$

$$= \int \frac{4x}{(x+1)(x^2+1)}\, dx = \int \frac{A}{x+1} + \frac{Bx+C}{x^2+1}\, dx$$

$$4x = A(x^2+1) + (Bx+C)(x+1)$$

① let $x = -1$

$-4 = 2A \Rightarrow A = -2$

② let $x = 0$

$0 = -2 + C \Rightarrow C = 2$

③ let $x = 1$

$4 = 2A + (B + C)2$

$4 = -4 + 2B + 4 \Rightarrow B = 2$

$$= \int \frac{A}{x+1} + \frac{Bx+C}{x^2+1}\,dx$$

$$= \int \frac{-2}{x+1} + \frac{2x+2}{x^2+1}\,dx$$

$$= -2\ln|x+1| + \int \frac{2x+2}{x^2+1}\,dx$$

Let $u = x^2+1$; $\frac{du}{dx} = 2x$ ⇒ $dx = \frac{du}{2x}$

$$+ \int \frac{2x}{x^2+1} + \frac{2}{x^2+1}\,dx$$

$$-2\ln|x+1| + 2\ln|x^2+1| + \tan^{-1}|x^2+1| + C$$

$\int \left|\frac{2x+2}{x^2+1}\right|$

$\frac{du}{u}$

$u+1$

29. $\int \frac{x^3 + 4x + 3}{x^4 + 5x^2 + 4}\,dx$

$= \int \frac{x^3+4x+3}{(x^2+4)(x^2+1)}\,dx$

↓ (x^2+4) 和 x^2+1 都是 complex roots

$= \int \frac{Ax+B}{x^2+4} + \frac{Cx+D}{x^2+1}\,dx$

$x^3+4x+3 = (Ax+B)(x^2+1) + (Cx+D)(x^2+4)$

$= Ax^3 + Ax + Bx^2 + B + Cx^3 + 4Cx + Dx^2 + 4D$

$= (A+C)x^3 + (B+D)x^2 + (A+4C)x + (B+4D)$

Comparing coefficient

$A+C = 1$
$B+D = 0$
$A+4C = 4$
$B+4D = 3$

$\Rightarrow$ $A = 0$, $B = -1$, $C = +1$, $D = 1$

$= \int \frac{-1}{x^2+4} + \frac{x+1}{x^2+1}\,dx$

$= -\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + \int \frac{x}{x^2+1} + \frac{1}{x^2+1}\,dx$

let $u = x^2+1$; $\frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$

$= -\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + \frac{1}{2}\ln|x^2+1| + \tan^{-1}(x^2+1) + C$

31. $\int \frac{x+4}{x^2+2x+5}\,dx$

$= \int \frac{x+4}{(x+1)^2+4}\,dx$

let $u = x+1 \Rightarrow x = u-1$

$dx = du$

$= \int \frac{u+3}{u^2+4}\,du$

$= \int \frac{u}{u^2+4} + \frac{3}{u^2+4}\,du$

let $w = u^2+4$; $\frac{dw}{du} = 2u \Rightarrow du = \frac{dw}{2u}$

$= \frac{1}{2}\int \frac{1}{w}\,dw + \frac{3}{2}\tan^{-1}\left(\frac{u}{2}\right)$

$= \frac{1}{2}\ln|w| + \frac{3}{2}\tan^{-1}\left(\frac{u}{2}\right) + C$

$= \frac{1}{2}\ln|u^2+4| + \frac{3}{2}\tan^{-1}\left(\frac{u}{2}\right) + C$

$= \frac{1}{2}\ln|x^2+2x+5| + \cdots$

35. $\int_0^1 \frac{x^3 + 2x}{x^4 + 4x^2 + 3}\,dx$

let $u = x^4 + 4x^2 + 3$; $\frac{du}{dx} = 4x^3 + 8x = 4(x^3 + 2x)$

$dx = \frac{du}{4(x^3+2x)}$

when $x = 1$, $u = 8$

$x = 0$, $u = 3$

$= \frac{1}{4}\int_3^8 \frac{1}{u}\,du$

$= \frac{1}{4}[\ln|u|]_3^8 = \frac{1}{4}(\ln 8 - \ln 3) =$

37. $\int \dfrac{5x^4 + 7x^2 + x + 2}{x(x^2 + 1)^2}\,dx$

↳ repeated roots + complex roots

$$= \int \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

51. $\int \frac{e^{2x}}{e^{2x} + 3e^{x} + 2}\, dx$

let $u = e^x$; $\frac{du}{dx} = e^x \Rightarrow dx = \frac{du}{e^x}$

$= \int \frac{u}{u^2 + 3u + 2}\, du$

$= \int \frac{u}{(u+2)(u+1)}\, du$

$= \int \frac{A}{u+2} + \frac{B}{u+1}\, du$